Rank two projective geometries: classification and corrections

Human Compression Project

Abstract

We give a complete treatment of the mathematical claims in the fixed published Part II article and its retained appendix. The published classification omits affine configurations over subfields. Its convention identifies the two parabolic subgroups without ordering them, which also changes two families of conjugacy counts. Four stated full graph groups require correction. The surviving configurations, their class counts, their local arc lengths and the remaining full group assertions follow from the proofs collected here. We distinguish these corrections from empty parameter cases, preliminary slips, classical premises and the contents missing from the retained appendix edition.

1 The two originals and the meaning of completion

The first original, designated L063, is Buekenhout, De Saedeleer and Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea 6 (2013), 365–388. It has 24 pages. The second, L064, is a retained 21-page appendix edition beginning with the proof of Lemma 12 and ending with Proposition 30. They are two separately registered originals. Shared arguments are included once in this collection, while the claim ledgers identify each original and its actual wording separately.

A complete treatment of a false assertion is its explicit disproof, not an attribution of the corrected assertion to the source. The omitted affine family was already established in the earlier accepted component; it is reused here. The present completion adds the proofs needed for the surviving assertions. It does not record another principal refutation of either original. In particular, the source's four impossible square-field rows are empty conditional cases, rather than four numerical counterexamples.

Our scope is the mathematical classification and its stated consequences. Historical computer timings and statements that particular examples were checked in MAGMA are attributed to their authors; those historical runs are not reproduced. The classification does not depend on them. The authors' earlier constructions, the Biggs–Hoare and Wong graphs, and the classical generalized quadrangle retain their original attribution.

2 A common proof structure

Put $G = \mathrm{PSL}(2, q)$, and write $\{M, L\}$ for two distinct proper parabolic subgroups, with $B = M \cap L$. We use the unordered convention in Section 2 of L063. Conjugacy means an orbit under G; isomorphism in that source means an orbit under $\mathrm{Aut}(G)$. This latter convention concerns the distinguished action and does not assert that every abstract graph isomorphism is induced by $\mathrm{Aut}(G)$.

The coset construction gives one object for each coset of M or L, with incidence defined by nonempty intersection. Every incident pair is translated to the standard pair, whose stabilizer

is B. Its local actions are M on M/B and L on L/B. Thus the required rank two conditions amount to

$$M \text{ or } L \text{ maximal in } G, \qquad |B\backslash M/B| = |B\backslash L/B| = 2.$$

The local actions are primitive and have degree at least two, proving firmness. The other parabolic is not contained in the maximal one, so the two generate G; this proves connectedness. For fixed M, ordered configurations are normalizer orbits on L. If the parabolic types agree, one must also identify the orbits obtained by exchanging them. This is the accepted common rank two argument from the Part I treatment, restated to fix the convention used throughout this collection.

The next step is to classify the possible two-transitive actions. For $d = [H : B]$, a point stabilizer is transitive on the other $d - 1$ points, so

$$d - 1 \mid |B|, \qquad |H| \leq |B|(|B| + 1).$$

The separately proved common action theorem combines this bound with the classical subgroup types, and treats the normal PSL subgroup of PGL explicitly. It supplies the projective, affine and small exceptional possibilities used in every subsequent application. We do not assume the source's representation proposition, whose cited proof was listed as being in preparation.

Proof component	Its role in the classification
Two-transitive coset actions in dimension two	Complete candidate lists for the local parabolics and intersections.
Subfield planes and unordered pairs of parabolics	Affine planes, their conjugacy and Frobenius orbits, and the odd-characteristic unordered count.
Counting configurations above a cyclic subgroup of order three	All seven Table 2 rows, including the corrected equal-type inner count.
Exceptional parabolics from their common normalizers	Tables 1 and 3 exceptional counts and the empty square-field cases.
Subfield parabolics and affine overgroups	Complete even, odd and quadratic subfield alternatives, including the small fields.
Local arc transitivity from small stabilizers	Every local arc column in Tables 1–6, including the nine Table 9 crossreferences.
The three graph group components	Full groups for the S_4, S_4 families, the S_4, D_{16} subdivisions, and the $q = 9$ quadrangle.

The subgroup classification of Dickson, Moore and Suzuki is an explicit classical premise, including the stated maximal subgroups, subfield embeddings, normalizers and fusion under PGL. The order of PSL and its action on a projective line are classical as well. The full graph group proofs additionally use Tutte's bound of 48 for a vertex stabilizer in a finite connected cubic arc-transitive graph, under its actual hypotheses. None of these premises is a retained version of the target Part II geometry classification. The accepted Part I treatment is used only where a maximal parabolic belongs to its Borel or dihedral scope.

3 The corrected classification and numerical columns

The complete action list separates the maximal parabolics of the Part II problem into A_4, S_4, A_5, proper PSL subfield groups and quadratic PGL subfield groups. For each allowed intersection,

the common normalizer proofs count actual configurations, rather than inferring their number merely from subgroup conjugacy. The exceptional and small field arguments supply all finite branches. The subfield proof shows that there are no further branches.

For clarity, the changes to the printed numerical columns are the following. All other nonempty rows retain their source class counts.

1. In even characteristic, let $K = \mathbb{F}_s \subset F = \mathbb{F}_{s^m}$ with $s \geq 4$ and m prime. Beside $M = \mathrm{PSL}(2, K)$, take

$$B = U_K : K^*, \qquad L = U_V : K^*, \qquad K \subset V \subset F, \quad \dim_K V = 2.$$

   Every such V gives the required configuration. There are $(s^{m-1} - 1)/(s - 1)$ conjugacy classes, and the automorphism classes are exactly the Frobenius orbits on these subspaces. At $m = 2$ this is the single printed Table 4 row. The assertion that $m = 2$ is necessary is false. At $s = 4$ the accepted scalar formulas recover the Table 1 counts as well.

2. In Table 2 rows 6–7, write $m = (p - \epsilon)/6$, where $3 \mid (p - \epsilon)/2$. The printed $2m - 1$ counts ordered configurations. For unordered configurations the correct inner count is

$$m + \lfloor m/2 \rfloor.$$

   The automorphism count remains m. The proof identifies the overgroups with a regular dihedral normalizer quotient and then takes inversion orbits.

3. In Table 5, for $q = 3^m$ with m an odd prime, the inner count is $(3^{m-1} - 1)/2$, rather than $3^{m-1} - 1$. At $q = 27$ it is four rather than eight. The printed automorphism counts are retained: two at $m = 3$, and $(3^{m-1} - 1)/(2m)$ otherwise.

4. In the square-field part of Table 1, $q = p^2 \equiv -1 \pmod 5$ with p odd implies $q \equiv 9 \pmod{40}$. Rows 12–14 and 18 therefore have no admissible parameters. Rows 15–17 remain; the alternative $q \equiv -1 \pmod{40}$ in rows 16–17 is also empty.

The complete source tables have $18 + 7 + 9 + 5 + 2 + 3 = 44$ rows. This is a count of printed rows, not of distinct families or new theorems. Table 4 row 5 repeats Table 1 row 1; the $s = 4$ subfield case overlaps the A_5 discussion; four printed square-field rows are empty. The ledgers record these overlaps without adding source credit. The nine entries of Table 9 point back to those same rows.

The original local arc lengths are proved for each actual configuration. The added even-characteristic affine configurations have local arc length three by the same local calculation. The S_4, S_4, D_8 configuration has length four for the specified simple group, and the S_4, D_{16}, D_8 configuration has length seven. These statements about G are kept separate from the full graph groups.

4 Full graph groups

For a prime $p \equiv \pm 1 \pmod 8$, choose $\epsilon \in \{1, -1\}$ with $p \equiv \epsilon \pmod 8$, and put $d = (p - \epsilon)/8$. The full group of a graph with parabolics S_4, S_4 and intersection D_8 is

$$\mathrm{Aut}\, X \cong \begin{cases} \mathrm{PSL}(2, p) \times C_2, & d \text{ even}, \\ \mathrm{PGL}(2, p), & d \text{ odd}. \end{cases}$$

An explicit torus-normalizer calculation constructs the required type interchange. Tutte's bound then gives index at most four over G. Simplicity forces G to be normal, and the centralizer and the action on the two S_4 classes determine which of the displayed groups occurs. Thus the printed PGL identifications at $p = 17, 31, 47, 79$ are incorrect; the identifications at $p = 23, 41, 71, 73, 89$ are recovered. In all these prime cases the full graph action still has arc length four.

For $q = 9$, the graph has the 15 duads and 15 synthemes on six letters as its two types. Its symplectic realization proves that it is the generalized quadrangle of order $(2, 2)$. A displayed permutation interchanges types and is checked directly on all 45 edges. Together with the faithful S_6 action this attains the Tutte bound of 1440. The full group is therefore $\mathrm{P\Gamma L}(2, 9)$, with full arc length five, as the source states. The construction and the named graph are classical; the literal duality certificate is a verification of the proof.

For $q = 17, 31, 79, 97$, suppressing the degree-two vertices in the S_4, D_{16} graph gives a simple cubic graph. The suppression proof excludes parallel edges and uses the actual transitive action. Tutte's bound gives index at most two, while the outer automorphism exchanges the two S_4 classes and cannot preserve the required action. The full group is $\mathrm{PSL}(2, q)$, recovering all four stated groups.

5 Preliminary formulas and literal corrections

The elementary order obstruction above proves L063 Lemma 4.4. For Lemma 4.5, an abelian transitive permutation group has the same stabilizer at every point. Faithfulness makes that stabilizer trivial, so the action is regular. If it is two-transitive, its trivial point stabilizer can be transitive on the other points only when there is one other point. Thus the degree is two. The word "permutation" here includes faithfulness; a group with a nonfaithful abelian action would first be replaced by its image.

The elementary abelian orbit formulas in Lemma 4.1 need their field parameter to be specified. This also explains why the source correctly declines to give a number of conjugacy classes from those formulas alone.

Lemma 1. *Let $0 \neq V \subset \mathbb{F}_q$ be an additive subgroup of order p^m, and let*

$$K = \{a \in \mathbb{F}_q : aV \subset V\} = \mathbb{F}_{p^k}.$$

Put $A_F = \mathbb{F}_q^*$ in characteristic two and $A_F = (\mathbb{F}_q^*)^2$ in odd characteristic. In $G = \mathrm{PSL}(2, q)$,

$$\mathrm{N}_G(U_V) = U_{\mathbb{F}_q} : (K^* \cap A_F).$$

If $1 \neq D \leq K^* \cap A_F$, then

$$\mathrm{N}_G(U_V : D) = U_V : (K^* \cap A_F).$$

Consequently the orbit lengths are respectively

$$\frac{q^2 - 1}{e(p^k - 1)}, \qquad \frac{(q^2 - 1)p^{n-m}}{e(p^k - 1)}, \quad q = p^n,$$

where $e = 2$ if p is odd and n/k is even, and $e = 1$ otherwise.

Proof. The set K is closed under addition and multiplication and contains one. Every nonzero element sends the finite set V bijectively to itself, so its inverse also lies in K. Thus K is a field, and k divides both n and m. A normalizer must fix the unique fixed point of U_V, hence has affine

form $x \mapsto ax + b$. It normalizes U_V exactly when $a \in K^* \cap A_F$; the translation b is unrestricted. For the second group U_V is characteristic as its unique Sylow subgroup for the characteristic. Conjugating any scalar $c \neq 1$ in D gives translation $(1-c)b$, which lies in V exactly when $b \in V$, since $(1-c)^{-1} \in K$. The displayed normalizers follow. Divide $|G| = q(q^2-1)/\gcd(2, q-1)$ by their orders and use the square criterion for K^* in $\mathbb{F}_q$ to obtain the formulas. □

Thus the source's k is the degree of this multiplier field, not an arbitrary common divisor. Different V can have different such fields; the formulas determine each orbit length, not the number of orbits.

Three additional preliminary corrections are explicit.

1. The odd-characteristic single-class clauses of Lemma 4.1(5) must require $d > 2$ when referring to D_{2d}. At $d = 2$ the four-groups belong to item (6). For example, at $q = 5$ the former formula would give 15 four-groups, whereas A_5 has five: each is the subgroup of double transpositions fixing one of the five letters. This verifies the correction without an ambient search.

2. The $\mathrm{PGL}(2, s)$ index in Table 8 is

$$[\mathrm{PSL}(2, s^2) : \mathrm{PGL}(2, s)] = \frac{q(q^2-1)}{2s(s^2-1)}, \qquad q = s^2.$$

   The printed denominator omits the factor two. This follows directly from the two group orders.

3. The printed exhaustive list in Proposition 4.6 omits the degree-two and degree-four actions of a maximal S_4 at prime q. For instance $S_4 < \mathrm{PSL}(2, 7)$ has its action on S_4/A_4 of degree two and its faithful natural action on S_4/S_3 of degree four. Its standalone clause lists degree three only; the separate PGL clauses impose $q = s^2$. The common action proof includes all three S_4 actions with their correct kernels. Assertions about representations inside one fixed subgroup are distinguished from the two ambient embedding classes.

L064's proof of Lemma 24 also incorrectly excludes every $\mathrm{PGL}(2, 3^w)$ from $\mathrm{PSL}(2, 3^{2m})$ for odd prime m. Both $w = 1$ and $w = m$ give subfield embeddings. The required two-transitive overgroup exclusion is nevertheless true and is proved by the common action theorem; it does not use that auxiliary assertion. In L063's proof of Lemma 5.3, Steps 4–5 confuse the labels A_4 and $E_{16} : 3$. The total number of the A_4 subgroups in the stated characteristic-two family is $q(q^2-1)/12$, rather than $q(q-1)/12$: the additional factor $q+1$ counts their unique fixed projective points. The accepted coordinate proof gives the correct intermediate counts while recovering the lemma's final formula. In three displayed transporter formulas of L064 Proposition 30, on physical pages 19, 20 and 21, the denominator must be the actual ambient order $|\mathrm{PSL}(2, p^{2n})|$, not $|\mathrm{PSL}(2, 2^{2n})|$. The literal normalizer proofs give the stated final values without that transcription error. In Section 6 of L063, the second path in the A_5, D_{30} argument starts at type one, and the S_4, D_{24} argument must say D_{24}, not D_{18}, when acting on cosets of D_8. The full local proofs in the collection fix both slips.

6 Separate attribution and source accounting

L063's Theorem 1.1 and its concluding completeness assertion are false as printed because of the omitted affine family. The corrected proof retains the valid configurations and all valid class

counts, then replaces the incomplete subfield argument and the unordered counts. The local arc columns are recovered, and the full graph groups have the stated four corrections. Preliminary borrowed results are identified as premises with the exact qualifications above; we do not claim to have rederived the classical subgroup classification.

L064 contains proofs for Lemma 12, Proposition 16, Lemmas 17–19, Proposition 20, Lemma 21, Proposition 22, Lemmas 23–25, Proposition 26, Lemmas 27–29 and Proposition 30. Its Lemma 21 lists the affine type without the published lemma's restriction, but Proposition 22 still restricts the configuration to extension degree two. The respective dispositions are therefore different. Its Propositions 16 and 26 share the unordered-count corrections, and Lemma 24 has the auxiliary correction just discussed. Its other surviving assertions follow from the same explicit arguments as L063.

The fixed appendix does not contain the advertised entire A_5 count proof, the standalone published $q = 25/S_4$ lemma, or Section 6's claimed arc proofs on appendix pages 19–20. Those pages actually concern Proposition 30. We supply the required proofs for L063 and do not invent missing L064 content or assign it extra claims. This is an edition boundary, not a statement that no other appendix edition ever existed.

The project contributions are the shared coordinate, transporter and normalizer arguments; explicit unordered orbit corrections; complete local stabilizer calculations; and the proofs and certificates identifying the full graph groups. Where an argument reuses a source proof, such as the fourth-arc obstruction in the S_4, D_{24} case, that attribution remains in its component. Distinct review records also disclose the collaborative contributions to the cubic group bounds and the Frobenius argument. Exact source ledgers and input bindings accompany the collection. No compression ratio, new open-conjecture answer or Lean verification is claimed in this completion. Those tasks remain outside the present intake stopping boundary.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Math. Contemp. **6** (2013), 365–388.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition.

[3] J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part I*, Ars Math. Contemp. **3** (2010), 177–192.